

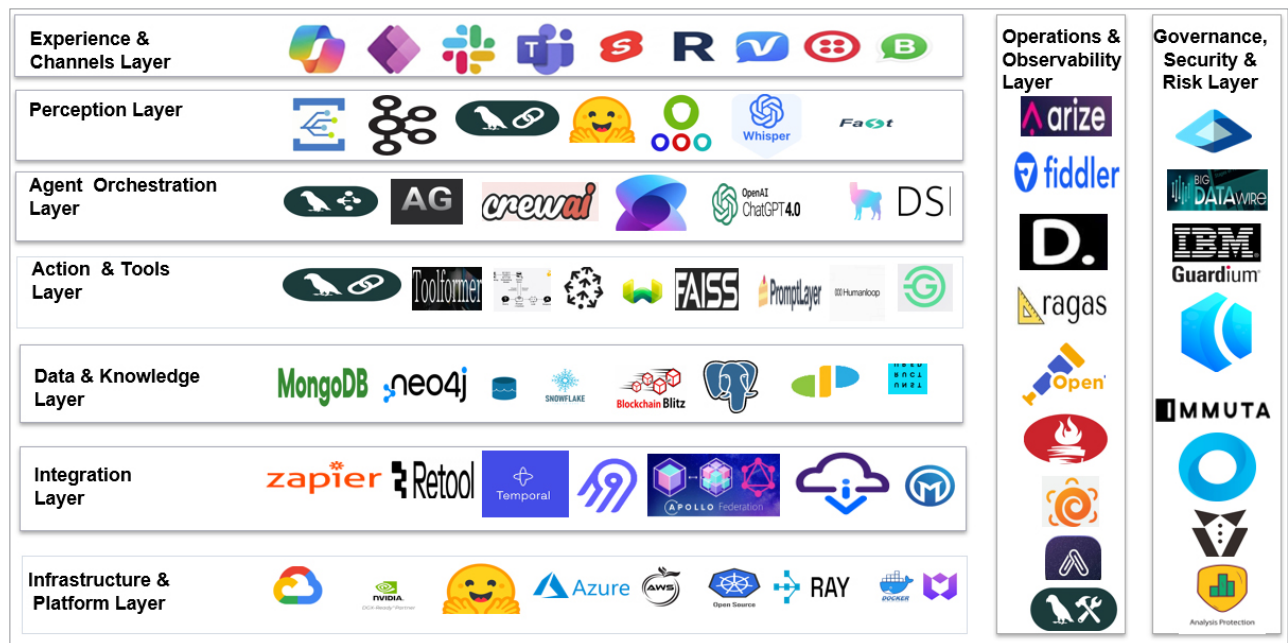


Figure 3: Tools usage across different layers of the agentic AI reference architecture

Open source tools for agentic AI architecture

This section briefly outlines open source tools used in the different layers of agentic AI architecture.

Experience and channel layer

- **Streamlit:** This framework facilitates rapid prototyping by allowing developers to create Python-based web interfaces for machine learning applications without deep frontend expertise.

Perception layer

- **Kafka:** This serves as the real-time event backbone, streaming data from various sources to the agent.
- **Hugging Face Transformers:** Provides pre-trained models for natural language processing (NLP) to interpret text data.
- **OpenCV:** A comprehensive toolkit for computer vision, allowing agents to process and understand visual inputs.
- **Whisper:** Specifically handles speech-to-text conversion, enabling voice-activated agentic workflows.
- **FastAPI:** High-performance web framework used to build APIs that ingest data and serve model predictions.
- **LangChain input parsers:** Framework that normalises varied input formats into structured data types that agents can process consistently.

Agent orchestration layer

- **LangGraph:** Framework that enables complex, stateful agentic workflows using directed acyclic graphs (DAGs).
- **AutoGen:** Specifically designed for building multi-agent

systems, where agents can converse to solve tasks.

- **CrewAI:** Manages role-based multi-agent flows, assigning specific 'personas' and tasks to different agents.
- **Semantic Kernel:** This integrates LLMs with conventional programming, focusing heavily on agent memory and planning capabilities.
- **LlamaIndex:** This acts as the interface between data and agents, ensuring the LLM can query and reason over private data.
- **DSPy:** Framework for a programmatic approach to optimising LLM prompts and weights, moving away from manual prompt engineering towards declarative programming.

Action and tools layer

- **LangChain:** Provides a standard interface for agents to call external functions, APIs, or scripts.
- **Toolformer:** Model trained to decide which tools to call and when, in a self-supervised manner.
- **RAG pipelines:** These implement retrieval-augmented generation to fetch relevant context before generating a response.
- **Weaviate:** Vector database used to store and retrieve high-dimensional data for RAG.
- **FAISS:** Library optimised for efficient similarity search and clustering of dense vectors.
- **Guardrails.ai:** Framework that ensures safety and reliability by validating agent outputs against specific quality and security constraints.